

More power to your elbow

Guy Kewney explains a new linking concept that could make a computer user's dreams come true

THE oldest dream of computer users running out of power on the most expensive machines they can afford is that by buying two of them, they might get twice the power. That dream may be closer to reality than the industry knows if Tasc Systems, a start-up company, achieves its second linking.

The concept has already attracted the eye of an unnamed British chip designer and Chuck Priddy, the American known as the father of the personal computer for his invention of the Commodore Pet.

Mr Priddy is helping Tasc will be at the heart of its new project linking "clusters" of processing units to provide enormous power at low cost.

Tasc has developed the core of a parallel operating system. Parallel processing has become a buzz word before, but has never overcome the two biggest problems of parallel working.

First, Tasc — the name of the operating system as well as the company — can ensure that the person writing the software does not have to know how local power processors are being used. Many parallel systems assume you are using all of the processors in a network. This can be made to work quickly, but it prevents anybody from adding more work to the computer until the first job stops.

Second, the system does not

really care which processor it is running on. The linking idea is it is perfectly feasible to have a multi-processor system with one Intel 80386, one Motorola 68000, three 586 chips from Intel, and a Sun Sparc.

Most parallel systems must be homogeneous. Tasc doesn't need to have a specific chip to

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work. The way it operates would normally be avoided because it would be too slow. It makes each chip in the network pretend to be a standard 386-processor called the VP, or virtual processor, and into each VP it loads a small program called Ales, which can run Tasc routines.

The founders have demonstrated the speed of the code by running routines under Tasc which assume they run on the Intel 80386 Transputer than the Transputer is capable of going when programmed in its own

machine code. What makes the claim seem incredible is that the software needed to run this will fit into less than 64K bytes of memory.

To put that into perspective, the IBM RS-6000 system announced last week needs at least eight million bytes of memory just for the operating software before it even begins to perform any useful applications.

The size of Tasc was developed by Chris Hensley, a programmer, as part of a project to create a computer games console. The power of his invention was illustrated recently when two games, *Megamind* and *Warhammer*, hit the market. Both are "arcade" games running on the Atari ST family, but display a level of activity and speed which could simply be impossible on a machine as restricted as the ST.

They key to his invention, according to Mr Hensley's partner, Mike Spicer, is that the program can locally do not need to know how the data is being processed. It is enough to define the data properly.

Mr Spicer said: "The technical term for this operating technology is 'atomic' and it is normally regarded as only theoretically possible. The key difference is that we don't write programs. We define data in terms of what will happen to it, and the operating system does that for you."

A test version of Tasc has been developed for the Intel 80386 Transputer, a machine which tends itself to be being issued together into multi-chip systems of enormous power.



There is no need to use Transputers. "Any processor that has a method of passing messages and data from one processor to another will do," Mr Spicer said.

He added that the basic development for the program could be ready before the end of this year. However, it will be some time before anything based on this technology appears in the shops.

For all its apparent advantage in terms of speed, most

computer users don't notice the prospect of creating their own programs, and prefer to buy known, low-cost packages with the software shop counter.

Mr Spicer believes that a system capable of running Tasc could be built to run standard software like the IBM-family software *Wordstar* and *Lotus 1-2-3*, and still be competitive in price with today's standard machines.

But it could be even that years before such machines are available generally at that sort of cost.

IBM abandons an unexciting convention

FOR THE first time in the memory of the computer world, IBM has abandoned its tradition of producing wondrous hardware and good-enough software.

Instead, it has issued a direct challenge to Intel, the world's leading maker of computer processors, and to Microsoft, the designer of most people's operating software. The IBM message is that it intends putting up a fight.

Last week IBM launched a range of work stations — personal and office computers — based on its own silicon technology. They are a range of computers that challenge the best of the world's silicon.

For the executive with suddenly disposable cash, the cheapest new Powerstations, priced from \$4,995, will do everything that IBM's bigger personal computers will do — but many times faster.

Overseen in the computer world was generally expected IBM to follow its tradition. Normally, the introduction of a new machine in this context would be a bit like one of IBM's recent gaps — a range with the safety wire of the 486, the 386, the 3090, the 3081, the 3080, the 3080 — 430 and finally, with a grand announcement, the 3090 — 530.

IBM's normal personal computers, however, are sold by a different division. They come as "PC" models, rising up to the mainframe and the mainframe.

The biggest of these is the recently announced model 70 with an 486-like chip, known simply as the 7000 (R).

IBM's product manager was asked why the cheaper of these machines appeared so superior to the model 70. "Well, really, it is a 'better' and the better offered."

some models are actually cheaper.

Officials of the group say we cannot claim as the representative that they were challenging Intel, which makes the 80386 chips and 80486 chips that power the IBM personal computers.

To prove this was not the case, they said that the Intel chips are well-contained, whereas the 85 system is one even piece of silicon. This is a claim that is a release.

An engineer invited to the launch said: "By the end of 1991, that processor will be on a single chip, and for the price, it already offers more performance than any other."

"IBM may not be planning to dump Intel, but it has certainly made a clear that it prepared to challenge them in new technology performance," he said.

IBM's obsession with naming its computers by number did trap its second research in the London launch. All the Powerstation machines are known by four-digit identifiers beginning with 70.

So first is the 7021 — 120, then the 7021 — 120, 530, 540, and finally the 7021 — 430, the 7080 — 430 and finally, with a grand announcement, the 7080 — 530.

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